

Experiment No-02

Aim: Write an Assembly language program to find largest number among ten numbers in memory and store the result into memory.

Software Used: Keil uVision5

Program:

```
                AREA largest, CODE, READONLY
                EXPORT __main

__main          MOVS R5, #09
                LDR R0, =0x1FFFF000
                LDRB R1, [R0]

UP              ADDS R0, #01
                LDRB R2, [R0]
                CMP R1, R2
                BHI L1
                MOV R1, R2

L1              SUBS R5, #01
                BNE UP

                ADDS R0, #01
                STRB R1, [R0]
                END
```

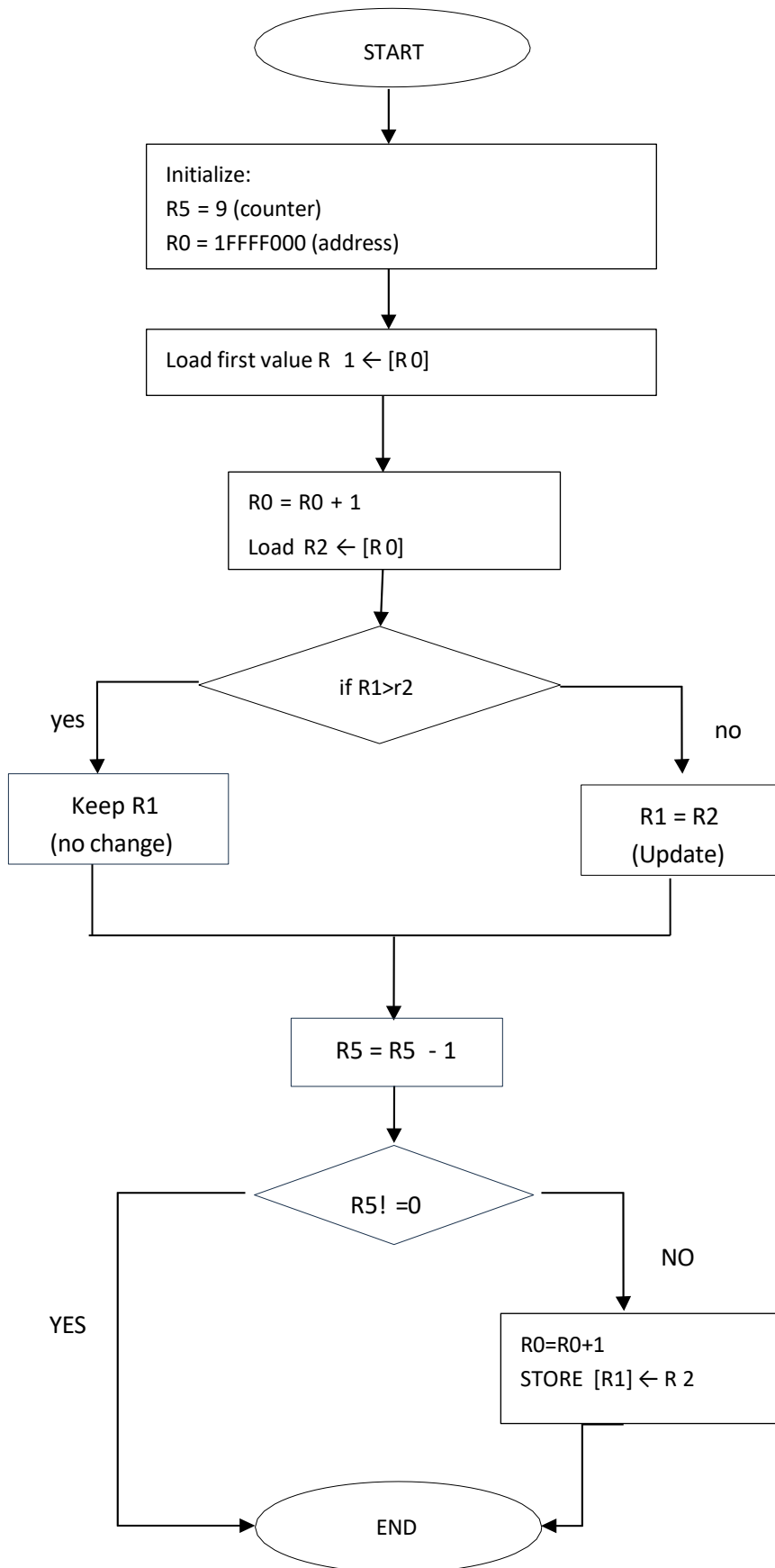
Program Screenshot:

The screenshot displays the uVision IDE interface with the following components:

- Registers:** A list of registers (R0-R15, xPSR) and their values. R15 (PC) is highlighted with a value of 0x000012C.
- Disassembly:** A window showing the assembly code for file2_exp2.s. The code includes instructions like ADDS, LDR, and STRB. The current instruction being executed is ADDS R0, #01.
- Command:** A window showing the command line and the error message: "Running with Code Size Limit: 32K", "Load 'C:\\Users\\cse\\Desktop\\b1_17_B_esd\\LAB3_EXP2\\Objects\\LAB3_FILE1_EXP2.axf'", "BS \\LAB3_FILE1_EXP2\\file2_exp2.s\\4", and "*** error 65: access violation at 0x40048100 : no 'write' permission".
- Memory:** A window showing the memory dump starting at address 0x1FFF000. The dump shows a sequence of 0x00 values.

The status bar at the bottom indicates the simulation is running, with a time of 0.00000892 sec and a CPU frequency of 14 C:10.

Flowchart:



Results and Conclusion:

Output:

Registers	
Register	Value
Core	
R0	0x1FFFF00A
R1	0x00000080
R2	0x00000023
R3	0x00000000
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x1FFFF100
R14 (LR)	0x000000C7
R15 (PC)	0x0000012C
xPSR	0x01000000
Banked	
System	
Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	107
Sec	0.00000892

Memory:

Memory 2	
Address:	0x1FFFF000
0x1FFFF000:	12 23 32 78 45 78 80 59 68 23 80 00
0x1FFFF01C:	00 00
0x1FFFF038:	00 00
0x1FFFF054:	00 00
0x1FFFF070:	00 00
0x1FFFF08C:	00 00

10 Numbers are compared and the output is available in Register R1 and Memory.